Work-energy lab

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Compare measured work with kinetic-energy change.

SUCCESS CRITERIA - I can...

1. Find work from force-displacement area.
2. Calculate kinetic energy from measured mass and speed.
3. Discuss disagreements without inventing measurements.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

A 3 N friction force opposes a 5 m displacement. Find work, including sign and units.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $W = F_{parallel}\, d$; $W_{net} = \Delta K$.
Constant force for the first equation; otherwise use signed force-displacement area. Use the approved cart setup and record actual trials separately.

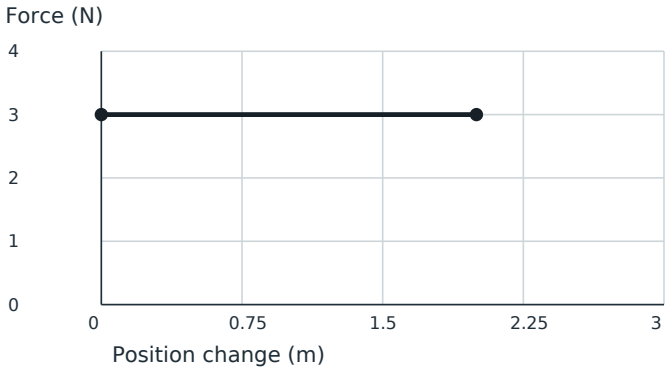
Worked reasoning: Supplied measurements give net work 12 J and kinetic-energy change 11 J. Their 1 J difference could reflect uncertainty or omitted work; it is evidence to investigate.

Your turn: Draw two bars for these measurements and label one check of the experiment.

Marzano check - circle: ① ② ③

03 Look and explain

The supplied force-displacement graph stays at 3 N over 2 m. Calculate work.



Marzano check - circle: ① ② ③

04 Build the solution

A 2 kg cart changes speed from 3 to 2 m/s. Find initial and final K, then ΔK. Compare with net work 3 J.

Marzano check - circle: ① ② ③

05 Use the idea

Create a trial table with mass, initial/final speed, work and ΔK. Name two causes of disagreement that deserve checking.

Marzano check - circle: 1 2 3

06 Choose a method

Two equal 4 kg carts approach at 6 m/s in opposite directions and stick. Find their common final velocity and explain.

Marzano check - circle: 1 2 3

07 Exit check

Why must the work and speed change refer to the same start and end positions?

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Two trials have differences 1 J and 5 J, each with possible difference uncertainty ±2 J. Which is consistent with zero under those bounds?

Marzano check - circle: 1 2 3